

Thresholds, Risk and Settlement: A Game-Theoretic Model of Last-Round Draws in Chess Tournaments

Ambrose Birtwistle

October 2026 — working draft

Abstract

In the final round of a chess tournament, players choose how much risk to take, and the prizes, title norms and rating consequences they face are step functions of the final score. I model a final-round pairing as a two-player game in which each player either *settles* for a draw or *fights* for a decisive result, and I show that the equilibrium depends on a single statistic of each player's payoff: the share ρ_i of the payoff step that lies below the draw. A mutual-settlement region exists if and only if $\rho_1 + \rho_2 > 1$, so draw agreements require aggregate concavity of payoffs around the current score and cannot arise with linear payoffs and no effort cost. When fighting is costly and decisiveness is complementary in the two players' efforts, settling and fighting are both self-enforcing conventions: draw collusion needs no side payments, only a focal point. The model yields testable implications for draw frequency, game length and engine-measured risk as a function of the standings before the last round.

1 Introduction

Chess differs from most tournament sports in one respect that matters for incentives: a game has three outcomes, and either player can often steer the game toward the middle one by simplifying the position. When prizes or title norms are awarded at score thresholds, the middle outcome can be worth nearly as much as a win to one player and nearly as little as a loss to another. The consequences are familiar to organisers, who have introduced rules such as bans on early draw offers and three-point scoring for wins, and to the Qualification Commission of FIDE, which reports having investigated tournaments where results were manipulated to obtain titles or norms and lists short draws among its indicators (FIDE Qualification Commission, 2024).

The existing economic literature on strategic draws is, to my knowledge, mostly empirical or simulation-based. The model here is complementary: it characterises when threshold-type payoffs produce mutual settlement, when they produce mutual risk-taking, and when both are equilibria.

2 Related literature

The closest empirical paper is Moul and Nye (2009), who use post-war tournament data to test whether Soviet players acted as a cartel by drawing among themselves, and conclude that the pattern of draws is consistent with collusion and inconsistent with competition. Their paper is a statistical test for a specific group and period; it does not derive how the shape of the prize function drives the choice between settling and fighting.

Recent operations-research work studies chess tournament design. Csató and Sziklai (2026) simulate Swiss-system team tournaments to quantify the trade-off between ranking accuracy and the probability of a stakeless last round, and note that match-point scoring in the Chess Olympiad creates situations where a draw on one board suffices for the team; they call for empirical study of how extra points for winning affect behaviour. Csató and Krumer (2026) document that in Swiss tournaments with an odd number of rounds, players with an extra game with the white pieces score more. These papers concern design and fairness rather than individual strategic choice.

More generally, tournament theory has shown that contestants may rationally shirk or even lose when future stages reward it (Vong, 2017; Krumer et al., 2023). Stakeless last-round matches have been analysed in football (Chater et al., 2021; Csató et al., 2024; Csató and Gyimesi, 2026), and Feddersen et al. (2023) find that betting markets anticipate effort adjustments at the end of the season. I am not aware of a chess model with a continuous prize step function and a draw outcome of the kind studied below, but I have not conducted a systematic search, and this should be checked before any claim of novelty.

3 The model

3.1 Setup

Two players $i \in \{1, 2\}$ are paired in the final round. Player i has current score s_i . Each chooses an action $a_i \in \{S, F\}$: *settle* (play to reduce variance and accept a draw when available) or *fight* (play for a decisive result). Fighting costs $c_i \geq 0$ in effort or preparation. Write $a = (a_1, a_2)$.

Assumption 1 (Decisiveness). The probability that the game is decisive is $D(a) \in (0, 1)$, strictly increasing in each player's action: $D(F, a_2) > D(S, a_2)$ and $D(a_1, F) > D(a_1, S)$.

Assumption 2 (Conditional win probability). Conditional on a decisive result, player 1 wins with probability $q \in (0, 1)$ and player 2 with probability $1 - q$, independently of a . Write $q_1 = q$ and $q_2 = 1 - q$.

The parameter q is where the rating difference enters; for instance it can be calibrated from a rating-based model of win, draw and loss probabilities (Elo, 1978; Milvang, 2015). Holding q fixed across actions is a simplification discussed in Section 7.

Assumption 3 (Payoffs). Player i 's payoff from the tournament is $u_i(x)$, weakly increasing in the final score x . A draw adds $\frac{1}{2}$ and a win adds 1 to s_i . Define

$$u_{i0} = u_i(s_i), \quad u_{ih} = u_i(s_i + \tfrac{1}{2}), \quad u_{i1} = u_i(s_i + 1),$$

$$U_i^+ = u_{i1} - u_{ih} \geq 0, \quad U_i^- = u_{ih} - u_{i0} \geq 0.$$

Whenever $U_i^+ + U_i^- > 0$ define

$$\rho_i = \frac{U_i^-}{U_i^+ + U_i^-} \in [0, 1].$$

The statistic ρ_i is the share of the payoff step lying *below* the draw: ρ_i near 1 means that a draw already secures almost everything and a win adds little (a concave region), while ρ_i near 0 means that only a win matters (a convex region); $\rho_i = \frac{1}{2}$ corresponds to linear payoffs.

3.2 Expected payoff

Lemma 1. For $i \in \{1, 2\}$,

$$\text{EU}_i(a) = u_{ih} + D(a)g_i - c_i \mathbf{1}\{a_i = F\}, \quad g_i = q_i U_i^+ - (1 - q_i)U_i^-.$$

Moreover $g_i > 0 \iff q_i > \rho_i$, and $g_i = 0$ if $U_i^+ + U_i^- = 0$.

Proof. The player wins with probability Dq_i , loses with probability $D(1 - q_i)$ and draws with probability $1 - D$, so $\text{EU}_i + c_i \mathbf{1}\{a_i = F\} = u_{i0}D(1 - q_i) + u_{ih}(1 - D) + u_{i1}Dq_i = u_{ih} + D[q_i(u_{i1} - u_{ih}) - (1 - q_i)(u_{ih} - u_{i0})]$. Finally $g_i > 0 \iff q_i(U_i^+ + U_i^-) > U_i^- \iff q_i > \rho_i$. $\square$

The number g_i is the net value to player i of adding variance: the gain on the upside weighted by the chance of winning, minus the loss on the downside weighted by the chance of losing.

Lemma 2 (Draw probability by profile). *The probability of a draw is $1 - D(a)$. By Assumption 1, $1 - D(S, S) > 1 - D(S, F)$, $1 - D(F, S) > 1 - D(F, F)$.*

4 Results

4.1 Benchmark: no effort cost

Proposition 1 (Dominant strategies). *Let $c_1 = c_2 = 0$ and $g_i \neq 0$. Then fighting is strictly dominant for player i if $g_i > 0$ and settling is strictly dominant if $g_i < 0$. Hence the game has a unique Nash equilibrium, in dominant strategies.*

Proof. By Lemma 1, for player 1, $\text{EU}_1(F, a_2) - \text{EU}_1(S, a_2) = g_1 [D(F, a_2) - D(S, a_2)]$, and the bracket is strictly positive by Assumption 1 whatever a_2 is. The sign is therefore that of g_1 . The argument for player 2 is identical. $\square$

Proposition 2 (Settlement and fighting regions). *Let $c_1 = c_2 = 0$ and suppose ρ_1, ρ_2 are defined and $q \notin \{\rho_1, 1 - \rho_2\}$.*

- (a) (S, S) is the equilibrium if and only if $1 - \rho_2 < q < \rho_1$. This interval is nonempty if and only if $\rho_1 + \rho_2 > 1$, and its length is $\rho_1 + \rho_2 - 1$.
- (b) (F, F) is the equilibrium if and only if $\rho_1 < q < 1 - \rho_2$. This interval is nonempty if and only if $\rho_1 + \rho_2 < 1$.
- (c) (F, S) is the equilibrium if and only if $q > \max\{\rho_1, 1 - \rho_2\}$; (S, F) if and only if $q < \min\{\rho_1, 1 - \rho_2\}$.

Proof. By Proposition 1 and Lemma 1, player 1 fights iff $q > \rho_1$ and player 2 fights iff $1 - q > \rho_2$, i.e. $q < 1 - \rho_2$. Intersecting the four sign combinations gives (a)–(c). The interval in (a) is nonempty iff $1 - \rho_2 < \rho_1$, and (b) iff $\rho_1 < 1 - \rho_2$. $\square$

Corollary 1 (Linear payoffs). *If $U_i^+ = U_i^-$ for both players (so $\rho_1 = \rho_2 = \frac{1}{2}$), the settlement region of Proposition 2 is empty: with no effort cost, linear payoffs never generate a mutually settled draw. Mutual settlement requires $\rho_1 + \rho_2 > 1$, that is, payoffs that are on aggregate concave around the current score.*

4.2 Threshold prizes

Prizes, title norms and rating-performance cutoffs are step functions of the final score. Let player i receive a prize $V > 0$ if and only if the final score is at least $s_i + r_i$, plus a small linear component $\varepsilon(x - s_i)$ with $\varepsilon > 0$ for the rating or tie-break value of each point:

$$u_i(x) = V \mathbf{1}\{x \geq s_i + r_i\} + \varepsilon(x - s_i).$$

Proposition 3 (Threshold requirements). *Let $r_i \in \{\leq 0, \frac{1}{2}, 1, > 1\}$ index how many additional points the player needs.*

- (i) *Draw suffices ($r_i = \frac{1}{2}$): $U_i^+ = \varepsilon/2$, $U_i^- = V + \varepsilon/2$, and $\rho_i = \frac{2V + \varepsilon}{2V + 2\varepsilon} = 1 - \frac{\varepsilon}{2(V + \varepsilon)}$. Player i settles iff $q_i < \rho_i$.*
- (ii) *Win needed ($r_i = 1$): $U_i^+ = V + \varepsilon/2$, $U_i^- = \varepsilon/2$, and $\rho_i = \frac{\varepsilon}{2(V + \varepsilon)}$. Player i fights iff $q_i > \rho_i$.*
- (iii) *Prize already secured or out of reach ($r_i \leq 0$ or $r_i > 1$): $U_i^+ = U_i^- = \varepsilon/2$, $\rho_i = \frac{1}{2}$, and player i fights iff $q_i > \frac{1}{2}$.*

When $\varepsilon/V \rightarrow 0$, in case (i) the player settles for every $q_i < 1$, and in case (ii) fights for every $q_i > 0$.

Proof. Direct substitution into the definitions of $U_i^\pm$ and ρ_i , and Lemma 1. For (i), $u_{i0} = 0$, $u_{ih} = V + \varepsilon/2$, $u_{i1} = V + \varepsilon$. For (ii), $u_{i0} = 0$, $u_{ih} = \varepsilon/2$, $u_{i1} = V + \varepsilon$. For (iii), V drops out of all differences. $\square$

Corollary 2 (Who settles). *If both players are in case (i) (for instance, both share first place with a draw), then $\rho_1 + \rho_2 > 1$ and the settlement region is nearly all of $(0, 1)$, with length $\rho_1 + \rho_2 - 1 \rightarrow 1$ as $\varepsilon/V \rightarrow 0$. If one player is in case (i) and the other in case (ii), then $\rho_1 + \rho_2 = \frac{2V + \varepsilon}{2V + 2\varepsilon} + \frac{\varepsilon}{2V + 2\varepsilon} = 1$: the leader settles and the challenger fights for almost every q . If both are in case (iii) the only stake is the small linear component and the conclusion depends on costs (Proposition 4).*

Remark 1 (Scoring rules). Under one point for a win and one half for a draw, the live requirements are $r \in \{\frac{1}{2}, 1\}$, and a draw suffices in one of two cases. Under a 3–1–0 rule the live requirements are $r \in \{1, 2, 3\}$ points, and a draw suffices in one of three. If live requirements were equally likely (an illustrative assumption, not an empirical claim), the share of draw-sufficient cases falls from $1/2$ to $1/3$, and the chance that both players are simultaneously in case (i), if independent, from $1/4$ to $1/9$. This is consistent with the stated motivation of three-point rules, but the rule also changes the values of $U_i^\pm$ themselves, so the comparison is only heuristic.

4.3 Effort costs and tacit coordination

Assumption 4 (Complementarity). Let $\Delta_1(a_2) = D(F, a_2) - D(S, a_2)$ and $\Delta_2(a_1) = D(a_1, F) - D(a_1, S)$. Then $\Delta_i(F) \geq \Delta_i(S) > 0$: the effect of one's own fighting on decisiveness is larger when the opponent also fights.

Proposition 4 (Costly fighting). *Let $c_i > 0$ and, for each i with $U_i^+ + U_i^- > 0$, define*

$$\rho_i^S = \rho_i + \frac{c_i}{\Delta_i(S)(U_i^+ + U_i^-)}, \quad \rho_i^F = \rho_i + \frac{c_i}{\Delta_i(F)(U_i^+ + U_i^-)},$$

so that $\rho_i \leq \rho_i^F \leq \rho_i^S$. Under Assumption 4, and generically:

- (a) *Fighting is strictly dominant for i iff $q_i > \rho_i^S$, and settling is strictly dominant iff $q_i < \rho_i^F$.*
- (b) *If $\rho_i^F < q_i < \rho_i^S$, player i plays a conditional strategy: fight if the opponent fights, settle otherwise.*
- (c) *If both players are conditional, the pure-strategy equilibria are exactly (S,S) and (F,F).*

In general, the pure equilibria are the profiles in which every dominant player plays her dominant action and every conditional player copies the opponent.

Proof. By Lemma 1, the gain to player i from fighting is $G_i(a_j) = g_i \Delta_i(a_j) - c_i$. If $g_i \leq 0$ then $G_i < 0$ for both a_j , so settling is dominant (and $q_i \leq \rho_i < \rho_i^F$). If $g_i > 0$, $G_i(F) \geq G_i(S)$ by Assumption 4. Hence fighting is dominant iff $G_i(S) > 0$, settling is dominant iff $G_i(F) < 0$, and otherwise $G_i(S) \leq 0 \leq G_i(F)$ (conditional). Solving $g_i \Delta_i(a_j) > c_i$ for q_i using $g_i = q_i(U_i^+ + U_i^-) - U_i^-$ gives the thresholds $\rho_i^{a_j}$. Part (c) follows because each player's best response is then to match the other's action. $\square$

Corollary 3 (Collusion as coordination). *When both players are conditional, settling and fighting are both self-enforcing. Mutual settlement does not require an agreement enforceable by side payments: it is a convention, and what selects it is whatever makes the draw a focal point for the two players (a shared team, national federation, club, friendship or reputation). Effort costs and complementarity widen the region in which settlement is an equilibrium relative to Proposition 2.*

5 Empirical implications

The model's observable object is the standings before the last round. They determine each player's requirement r_i (Proposition 3), hence ρ_i and the predicted regime.

- H1** The probability of a draw is higher when $\rho_1 + \rho_2 > 1$ than when $\rho_1 + \rho_2 < 1$ (Propositions 2 and Lemma 2).
- H2** Holding ratings fixed, the draw probability changes sharply at the score margin at which a draw starts to suffice (case (i)), because ρ_i jumps between cases (ii) and (i).
- H3** Within the settlement region, draws are more frequent between players who share an affiliation (Corollary 3), the channel studied empirically by Moul and Nye (2009) for a specific group.
- H4** Risk measured by an engine (for example the volatility of the engine evaluation along the game, or the loss in centipawns) is lower under mutual settlement and higher under mutual fighting.

The data requirement is tournament-level results with final-round pairings, scores, ratings and, ideally, prize or norm structures. Results sources such as chess-results.com are used in the Swiss-system literature (Csató and Sziklai, 2026); their terms of use should be checked before collection. Draw offers are generally not recorded in over-the-board data, so game length is the natural observable proxy.

6 Extension: draw offers as signals (sketch)

The static model treats the game’s decisiveness as an exogenous function of actions. A natural next step lets player 1 hold private information about the objective evaluation of the position, $\theta \in \{\text{good}, \text{bad}\}$, and lets her offer or withhold a draw. The opponent updates on the offer and accepts or refuses, and the payoff step ρ_i determines which types are willing to offer. This has the structure of a signalling game, as in a prior paper of mine on programmatic signalling, but I have *not* derived or verified any result for it here, and it is left as a direction for a second paper.

7 Limitations

The model is static with binary actions and takes q as independent of behaviour. If instead rating-implied expected scores were held fixed whatever the players do, a decisive game would shift q with D , and the region boundaries would move; the qualitative role of ρ_i would remain but Propositions 1–2 would need to be restated. Payoffs are reduced-form functions of the score and ignore tie-break systems, team incentives and the possibility that draws affect others in the standings. The model also covers a single pairing, so it ignores the interaction between simultaneous boards (relevant to prizes decided by several games).

Appendix: numerical check

I verified Propositions 1, 2–3 and 4 by brute-force search over pure-strategy equilibria for randomly drawn D , q , payoff steps and costs (several hundred thousand draws each, with random weakly increasing u_i including flat steps), and found no counterexample. This is a sanity check on the algebra, not a substitute for the proofs above.

References

- Chater, M., Arrondel, L., Gayant, J.-P., and Laslier, J.-F. (2021). Fixing match-fixing: Optimal schedules to promote competitiveness. *European Journal of Operational Research*, 294(2):673–683.
- Csató, L., Molontay, R., and Pintér, J. (2024). Tournament schedules and incentives in a double round-robin tournament with four teams. *International Transactions in Operational Research*, 31(3):1486–1514.
- Csató, L. and Gyimesi, A. (2026). Increasing competitiveness by imbalanced groups: The example of the 48-team FIFA World Cup. *European Journal of Operational Research*, 332(3):868–879.
- Csató, L. and Krumer, A. (2026). Swiss-system chess tournaments and unfairness. arXiv:2410.19333 (v3).
- Csató, L. and Sziklai, B. R. (2026). The efficacy-competitiveness frontier in the Swiss system. arXiv:2610.01997.
- Elo, A. E. (1978). *The Rating of Chessplayers, Past and Present*. Arco, New York.
- Feddersen, A., Humphreys, B. R., and Soebbing, B. P. (2023). Contest incentives, team effort, and betting market outcomes in European football. *European Sport Management Quarterly*, 23(3):605–621.

- FIDE Qualification Commission (2024). Policy on Ratings and Titles. <https://www.fide.com/2024/01/31/>.
- Krumer, A., Megidish, R. and Sela, A. (2023). Strategic manipulations in round-robin tournaments. *Mathematical Social Sciences*, 122:50–57.
- Milvang, O. (2015). Probability for the outcome of a chess game based on rating. Manuscript.
- Moul, C. C. and Nye, J. V. C. (2009). Did the Soviets collude? A statistical analysis of championship chess 1940–1978. *Journal of Economic Behavior & Organization*, 70(1–2):10–21.
- Vong, A. I. K. (2017). Strategic manipulation in tournament games. *Games and Economic Behavior*, 102:562–567.